

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Analysis of Training and Testing MSE

A neural network is trained to predict student scores.

Training MSE = 0.85

Testing MSE = 5.20

The **Mean Squared Error (MSE)** measures the average squared difference between the predicted and actual values. A **lower MSE indicates better model performance**.

Since the testing MSE (5.20) is much higher than the training MSE (0.85), the model performs well on the training data but poorly on unseen test data. This indicates that the model **does not generalize well**, and the most likely problem is **overfitting**.

Possible Reasons for the Difference

1. Overfitting (Main Reason)

- ✓ The neural network has learned the training data too accurately, including its noise and outliers.
- ✓ Instead of learning general patterns, it memorizes the training examples.
- ✓ As a result, it achieves a very low training error but a high testing error.

Evidence:

- Training MSE = 0.85 (very low)
- Testing MSE = 5.20 (much higher)

2. Insufficient Training Data

- A small dataset may not represent all possible student performance patterns.
- The network cannot learn generalized relationships.
- The model becomes sensitive to specific examples.

Effect:

- Poor prediction on new student data.

3 . Excessive Model Complexity

The neural network may contain:

- Too many hidden layers
- Too many neurons
- Too many trainable parameters

A highly complex model can memorize training samples instead of learning useful trends.

4. Poor Data Quality

The training data may contain:

- Missing values
- Incorrect student scores
- Duplicate records
- Noisy observations
- Outliers

These reduce the model's ability to generalize.

5. Distribution Difference

Training and testing datasets may represent different populations.

Example:

- Training data contains mostly average-performing students.
- Testing data contains many high- or low-performing students.

This causes poor performance during testing.

6. Lack of Regularization

If regularization techniques are not used, the model becomes overly flexible and fits every training point.

Examples:

- No Dropout
- No L1 Regularization
- No L2 Regularization

7. Too Many Training Epochs

If training continues for too long:

- Training loss keeps decreasing.
- Testing loss starts increasing.

This is another sign of overfitting.

8. Poor Feature Selection

Important factors affecting student scores may be missing.

For example:

- Attendance
- Previous grades
- Study hours
- Assignment marks
- Class participation

Without informative features, the model cannot make accurate predictions.

Suitable Corrective Measures

1. Increase Training Data

- Collect more student records.
- Use diverse data from multiple classes or years.
- More data improves generalization.

2. Apply Regularization

Regularization prevents the network from becoming too complex.

Methods include:

- **L1 Regularization:** Encourages sparsity by shrinking some weights to zero.
- **L2 Regularization (Weight Decay):** Penalizes large weights and keeps them small.
- **Dropout:** Randomly disables neurons during training, forcing the network to learn robust features.

3. Use Early Stopping

Monitor the validation loss during training.

- Stop training when validation error begins to increase.
- Prevents overfitting.

4. Reduce Network Complexity

Simplify the architecture by:

- Reducing hidden layers.
- Reducing neurons.

- Removing unnecessary parameters.

Simpler models usually generalize better.

5. Data Cleaning

Improve data quality by:

- Removing duplicate records.
- Filling missing values.
- Eliminating outliers.
- Correcting incorrect labels.

6. Normalize or Standardize Data

Scale input features so they are on similar ranges.

Benefits:

- Faster convergence.
- More stable learning.
- Improved prediction accuracy.

7. Hyperparameter Tuning

Optimize:

- Learning rate
- Batch size
- Number of epochs
- Number of hidden layers
- Number of neurons
- Activation functions

Methods:

- Grid Search
- Random Search
- Bayesian Optimization

8. Cross-Validation

Instead of using a single train-test split, use **k-fold cross-validation**.

Benefits:

- More reliable performance estimate.
- Reduces bias due to a particular data split.

Comparison of Overfitting, Underfitting, and Good Fit

Model Type	Training MSE	Testing MSE	Interpretation
Good Fit	Low	Low	Learns patterns and generalizes well
Overfitting	Very Low	High	Memorizes training data; poor generalization
Underfitting	High	High	Model is too simple to learn patterns

In this problem:

- **Training MSE = 0.85 (Low)**
- **Testing MSE = 5.20 (High)**

This clearly indicates **overfitting**.

Final Conclusion

The neural network achieves an excellent training MSE of 0.85 but a much higher testing MSE of 5.20, indicating that it overfits the training data and fails to generalize to unseen student records. The gap may result from a complex model, insufficient or noisy training data, lack of regularization, excessive training epochs, or differences between the training and testing datasets.

To improve performance, use regularization (Dropout, L1/L2), early stopping, more and cleaner training data, cross-validation, feature scaling, hyperparameter tuning, and, if necessary, simplify the network architecture. These measures help reduce overfitting and improve prediction accuracy on new data.

Code of Conduct:

I B.Jahnavi / 192325043 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Jahnavi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided